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When Does On-Device Self-Training Help? Cross-Modal Teacher Signals, Random Probes and a Performance Guard for Smartphone-Movement Affect Estimation

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ABSTRACT Smartphone movement can estimate arousal and valence on the device, but a model trained on other people captures each user's *level* better than the *changes within* that user, and a few self-reports cannot fix this. We ask whether a dense but noisy teacher, facial expression from the front camera during screen use, can personalise a movement-only student model on the phone. We build the complete loop on Android (a 33-parameter student trained hourly from a blendshape teacher, random probes as the only ground truth, and a hysteresis guard) and replay 77 users of the K-EmoPhone data set through a simulator that reproduces the device's training rule and controls teacher accuracy, density and bias. The answer is a condition rather than a gain: with one week of data, none of six update rules, even with noise-free label copies, raises within-user AUROC above the population model (0.51 to 0.60 versus 0.64 for arousal); pooled AUROC rises (0.62 to 0.63–0.73) only because the student learns each user's level. The fixed-threshold guard detects a student 0.10 worse than the base with probability 0.76 within 14 days but reverts an equally good student 48 % of the time; no guard avoids this trade-off at three probes per day. Model-triggered prompts select more extreme moments and too few responses for per-user evaluation, justifying the random arm. The 32 features re-identify a user among 77 at most 7 % of the time, the 33 student weights at 82 %, so weights must stay on the device.

INDEX TERMS Affective computing, on-device learning, self-training, mobile sensing, experience sampling, performance guard, privacy, smartphone accelerometer.

I. INTRODUCTION

Smartphones record their owner's movement continuously through the accelerometer. A compact model that maps movement summaries to arousal and valence can run entirely on the device, which keeps a sensitive signal off the network and makes it available to applications that adapt content, pacing or notifications in real time. Two obstacles stand in the way. First, affect expressed in movement is individual: a model trained on other people transfers only partly, and the within-person variation that applications care about is exactly the part a population model captures least. Second, the obvious remedy, personalising the model with the user's own self-reports, is limited by how rarely people answer surveys. Three answers a day for a week give about twenty labels, far fewer than even a 33-parameter model needs to beat its prior.

This paper studies an alternative supply of labels. While the screen is in use, the front camera can estimate facial expression, and expression is a well-studied correlate of momentary affect. We treat expression-derived labels as a *teacher* for a *student* model that uses movement features only, so that the student keeps estimating when the camera is off. The teacher is dense (one label per minute of screen use) but noisy; survey answers are sparse but accurate. Whether self-training with the dense noisy signal can beat fine-tuning on the sparse accurate signal is the central question, and the answer depends on the teacher's accuracy and density. Because a self-trained model can also drift, we evaluate it against the population model on *random* probes that do not depend on the model's own output, and revert to the population model when the student falls behind.

We make the following contributions.

- **C1 (system).** A complete on-device loop on Android: minute-level movement features, a LightGBM population model, a MediaPipe-blendshape teacher, a LiteRT logistic student trained on the phone with a closed-form gradient, a contact gate, a two-arm prompting scheduler (model-triggered arm A, random arm B) and a hysteresis guard. One hourly update costs tens of milliseconds on a 2018 mid-range phone (Section VI-F).
- **C2 (conditions for self-training).** A replay simulator that runs 77 users of K-EmoPhone through the loop with controlled teacher accuracy p, density d and bias b, comparing the student with the fixed population model, ESM fine-tuning, per-user Platt scaling and an oracle teacher. The pre-registered hypothesis that a dense noisy teacher beats sparse accurate labels above some accuracy p^* is not supported in any of the 32 cells, and a stronger result constrains all approaches: with about one week of data none of the six learners we tested improves within-user discrimination over the population model, and the apparent gains are level calibration visible only in pooled AUROC (Section VI-B).
- **C3 (guard and selection bias).** A Bernoulli analysis of the hysteresis guard against SPRT, Bayesian and confidence-interval alternatives at three random probes per day (Section VI-C), and a measurement of how model-triggered prompts bias accuracy estimates (Section VI-D).
- **C4 (privacy).** Re-identification attacks on the 32 movement features with stronger attackers and smaller candidate sets, and an attack on the 33 student weights that a device could export (Section VI-E).
- **C5 (reproducibility).** All experiments are scripts in the repository; every number in this paper is produced by a script whose output file is cited next to it, and the simulator's training rule is tested against the LiteRT model's probabilities.

II. RELATED WORK

A. ON-DEVICE AND CONTINUAL LEARNING ON MOBILE DEVICES

Training on the phone itself, rather than only running inference there, has become practical in mainstream toolchains. TensorFlow Lite (now LiteRT) added on-device training in TensorFlow 2.7, exposing separate training and inference signatures in one model file and persisting weights through checkpoints so that each update round can resume from the previous one [1]. Work on memory-constrained training has concentrated on updating few parameters, since activation memory rather than parameter count dominates backpropagation cost: TinyTL freezes the weights and trains only biases and a lightweight residual module, cutting training memory by up to 6.5× [2], and LifeLearner combines meta-learning, rehearsal and compression to run continual learning on embedded platforms with a 178.7× memory reduction while staying within a few points of an oracle [3]. Federated learning moves the aggregation of such local updates to a server while keeping raw data on the device [4]; a large body of follow-up work addresses the fact that a single global model fits heterogeneous clients poorly and proposes personalized federated variants that keep part of the model client-specific [5]. In affect sensing specifically, Meegahapola et al. found that mood models trained across eight countries generalized poorly and that country-specific or partially personalized models performed best [6], which motivates local adaptation. This literature assumes that labelled examples arrive on the device. In our setting the update signal is a noisy proxy from another sensor and the model is a 33-parameter logistic regression, so the question is not whether training fits in memory but whether the signal is good enough to help.

B. MOBILE SENSING FOR AFFECT ESTIMATION

Passive smartphone and wearable sensing has been paired with in-the-wild experience sampling (ESM) to estimate momentary affect for over a decade. K-EmoPhone provides 77 participants, seven days, and 5,582 ESM responses with valence, arousal, stress and attention labels aligned to phone context and wearable physiology [7], and its successor EmoPhone extends the design to three yearly waves and shows that the best method depends on whether prediction is within-user, cross-user, or cross-wave [8]. Movement-only inputs carry affective information: Wampfler et al. reached a valence AUC of 0.83 from inertial signals recorded during typing on 82 users over ten weeks, with personalized models adding several points [9], and Woll et al. predicted momentary mood from 64 Hz accelerometry in 259 people, obtaining R^2 of about 0.38 at the hip and the weakest results at the wrist [10]. Taylor et al. treated each person as a task in a multitask model and reported large gains over a single population model for next-day mood [11]; Meegahapola et al. reported the same ordering across countries [6]. Fisher et al. showed that group-level associations frequently fail to describe the individuals in the group, so between-user accuracy and within-user accuracy are different quantities and should be reported separately [12]. We report both throughout. The personalization gains above were obtained with server-side retraining, self-reported labels, and weeks to months of data. Whether they survive a proxy label, on-device training, and about one week of data has not been measured; our K-EmoPhone replay is designed to answer this.

C. SELF-TRAINING, CROSS-MODAL TEACHER-STUDENT LEARNING, AND NOISY LABELS

Using one modality to supervise another is well established. Albanie et al. distilled a facial-expression network into a speech emotion model from unlabelled video, and reported that the student stayed below the teacher and below full supervision [13]. In the sensor domain, Ni et al. distilled a video action-recognition teacher into an accelerometer student that runs on the sensor alone at test time [14], and Tang et al. combined teacher-student self-training with self-supervision on inertial data, reporting F1 gains of up to 12 points and parity with ten times fewer labels [15]. Closest to our configuration, Onim et al. used iMotions facial-expression analysis as the ground truth for a wearable physiological regression model in 40 older adults, so that the camera is needed only during training [16]. Semi-supervised stress detection in the wild addresses the same label scarcity [17], and Noisy Student shows that self-training can improve a student when noise is injected deliberately and the student sees more data than the teacher [18]. Co-training formalizes why two conditionally independent views of the same target can bootstrap each other [19]; face and movement features are two such views of affect, which supports the design but requires each view to be individually informative. The learning-with-noisy-labels literature sets the conditions. Under the classification noise model, learning remains possible as long as the flip rate stays below one half, with sample complexity growing as $1/(1-2\eta)^2$ [20], and surveys catalogue how label noise degrades classifiers and which methods are robust [21]. Hence a teacher with accuracy p contributes information proportional to $(2p-1)$, and the pseudo-label density d needed to match a given amount of true labels grows rapidly as p approaches 0.5. Prior cross-modal work reports a single operating point; we instead treat teacher accuracy p and density d as controlled variables and map the region where self-training helps.

D. PERSONALIZATION, ESM BURDEN, AND SELECTION BIAS IN TRIGGERED PROMPTING

Dense self-report is the direct route to personalization, but it has costs. Reviews of mobile ESM document the design space of signal-, event- and interval-contingent prompting and the trade-off between label density and participant burden [22]. Eisele et al. showed experimentally that questionnaire length, more than prompt frequency, raised perceived burden and careless responding [23], and Reiter and Schoedel found in 592 participants that non-response is predictable from past compliance and context, so missingness is not random even under random prompting [24]. A more specific problem arises when the prompt is triggered by a sensor or a model. Lathia et al. quantified what they called contextual dissonance: sensor-triggered ESM changes the distribution of sampled behaviours relative to random sampling, and the bias depends on the trigger design [25]. A model that triggers prompts at moments it finds informative and is evaluated on those responses inherits that bias. The standard remedies are inverse-probability weighting, which reweights observations by the inverse of their sampling probability [26], and randomization of the sampling time itself, as in micro-randomized trials for just-in-time interventions [27]. We take the second route for evaluation (a random-time arm whose responses are the only ground truth) and use the first to quantify how far model-triggered responses depart from the random-time distribution.

E. PRIVACY OF MOVEMENT FEATURES AND MODEL PARAMETERS

Accelerometer data is not anonymous. Na et al. re-identified over 85% of adults and children in a national activity data set from 20-minute aggregated accelerometry after protected health information had been removed [28], Neverova et al. learned user identity from phone motion patterns across 1,500 users [29], and Kröger et al. review the range of inferences, from location and activity to identity and health, available from accelerometer streams [30]. Malekzadeh et al. reduced identity recognition below 7% while keeping activity recognition above 92% with an on-device transformation [31]. A second channel is the model itself. Membership inference recovers whether a record was in the training set from model outputs [32], and gradient or parameter updates in collaborative training leak attributes of the participants' data that are unrelated to the task [33]; differentially private SGD bounds this leakage at a cost in utility [34], and in affect recognition a multitask differentially private federated design has been used to trade identity leakage against stress accuracy [35]. For an on-device system that never uploads data, the remaining channel is the 33 parameters, which could leave the device through backups or diagnostics. Earlier work either anonymizes the features or protects the model; we measure both for the same summary features and hourly-updated weights.

III. PROBLEM SETTING AND NOTATION

A user u produces a stream of minute-level movement summaries $x_t \in \mathbb{R}^{32}$ (median and interquartile range over the past 30 minutes of 16 per-minute features, Section IV-A). Ground truth is an occasional self-report $y_s \in \{0,1\}$ (arousal above neutral; valence above neutral) at times s drawn by the random scheduler (arm B). The population model f_0 is trained on other users. The student g_θ is a logistic model on standardised x_t with 33 parameters, initialised from a population logistic seed θ_0. A teacher provides pseudo-labels $\tilde{y}_t$ on a subset of minutes with density d (fraction of minutes labelled) and accuracy $p = \Pr(\tilde{y}_t = y_t)$, possibly with class-dependent bias b. The device updates θ hourly by 100 full-batch gradient steps on the most recent 2,000 teacher-labelled minutes.

Evaluation uses two views of accuracy that behave differently under personalisation. **Pooled AUROC** ranks all probes of all users together and therefore rewards getting each user's *level* right. **Within-user AUROC** is computed per user and averaged, and rewards tracking *changes within a person*. Level can be learned from a handful of labels; tracking changes cannot. We report both throughout and treat within-user AUROC as primary (pre-registered in `docs/PREREG.md` before any simulation was run).

IV. SYSTEM DESIGN

A. MOVEMENT FEATURES AND POPULATION MODEL

The phone samples the accelerometer at the rate the OS delivers (about 50 Hz on the test device), interpolates to an 8 Hz grid and block-averages to 2 Hz, the rate at which the public data set and the device agree exactly (`src/s6_build_2hz.py`, `Features.java`). Each minute yields 16 orientation-free features of the gravity-removed magnitude (five distribution statistics, dominant frequency, spectral entropy, 0.1 to 1 Hz band power, five jerk statistics, per-second intensity mean and coefficient of variation, still fraction). The 30-minute summary is the median and IQR of each feature, 32 values. The population model is a LightGBM classifier with 50 trees of 3 leaves, trained on K-EmoPhone (Section V-A) and flattened to a 7.5 KB table for the device. Section VI-A shows that a 33-parameter logistic model on the same features is statistically indistinguishable from it, which is why the student can be that small.

B. TEACHER: FACIAL BLENDSHAPES DURING SCREEN USE

When the screen is on and the user holds the phone, the front camera runs MediaPipe Face Landmarker at 2 frames per second and keeps only the 52 blendshape coefficients; no image is stored. Per minute the app aggregates the mean smile coefficient (average of `mouthSmileLeft` and `mouthSmileRight`), the mean `jawOpen` and the fraction of frames with a detected face. The teacher emits a valence pseudo-label when smile exceeds θ_v and an arousal pseudo-label when smile plus jaw exceeds θ_a, provided the face-detection rate is at least 0.3. Thresholds are set per user as the 70th percentile of the first days' coefficients, or, once at least 20 random-probe answers exist, by maximising agreement with those answers (`src/s8_teacher.py`). This rule is deliberately simple and is the component whose validity we can least establish from public data (Section VII).

C. STUDENT AND ON-DEVICE TRAINING

The student is $g_\theta(x) = \sigma(w^\top z + b)$ with $z = (x - \mu)/s$, where μ, s are population statistics stored in the model. It is exported as a LiteRT model with `infer` and `train` signatures. Automatic differentiation produced an operator (`BroadcastGradientArgs`) that is not a built-in LiteRT operator, so the gradient is written in closed form: $\nabla_w = Z^\top(\sigma(Zw+b)-y)/n$, $\nabla_b = \mathbf{1}^\top(\sigma(Zw+b)-y)/n$, learning rate 0.05. Weights are read and written through `get_weights`/`set_weights` signatures because the checkpoint operators require the Flex delegate. The trainer keeps the most recent 2,000 teacher-labelled minutes and runs 100 steps once an hour when at least 50 are available (`OnDeviceTrainer.java`). The numpy re-implementation used by the simulator reproduces the LiteRT probabilities on the fixed probe batch to within 10^{-4} (`tests/test_loop_sim.py`).

D. CONTACT GATE

Estimation is suspended, but logging continues, when the device is charging, or when the screen is off and the still fraction of the past minute exceeds 0.9 (`ContactGate.java`). On K-EmoPhone the equivalent rule suspends 18.3 % of windows; arousal AUROC on suspended windows is lower (0.595 vs 0.623), although the participant-bootstrap confidence interval of the difference includes zero (+0.028 [−0.021, 0.080]).

E. TWO-ARM PROMPTING AND THE PERFORMANCE GUARD

Arm A prompts when the current arousal estimate differs from the estimate at the previous prompt by at least 0.25 and at least 30 minutes have passed. Arm B prompts three times a day at uniformly random times between 09:00 and 23:00, at least 30 minutes apart. Only arm B answers are used for evaluation and for the guard. The guard scores the population model and the student on every arm B answer (threshold 0.5), keeps a 14-day window, and once $n \geq 15$ reverts to the population model when the student's accuracy is more than 0.10 below it, re-enabling the student when it is at least 0.05 above (`DriftGuard.java`). Section VI-C evaluates these constants.

F. IMPLEMENTATION

Table 1 lists the device, runtimes, model files and tests of the implementation.

TABLE 1. Implementation of the on-device loop. Source: `docs/REPO_MAP.md`, `results/S8_NOTES.md`.

Component	Choice
Device	Samsung Galaxy Note9 (SM-N960N), Android 10; app targetSdk 29, minSdk 26, arm64
Runtime	LiteRT (org.tensorflow:tensorflow-lite 2.16.1), MediaPipe tasks-vision 0.10.35, CameraX 1.4.2
Training-side tooling	Python 3.11.9, LightGBM 4.7.0, scikit-learn 1.8.0, TensorFlow 2.19.0 (model export only)
Model files	population model 7.5 KB flattened trees; student `head.tflite` 9.4 KB; seed weights 2.5 KB text
APK	22.5 MB (MediaPipe and LiteRT included); no INTERNET permission
Tests	13 pytest (feature and label pipeline, golden log replay) + 31 JUnit (scheduler, guard, contact gate, student parity) + 2 simulator tests

V. EVALUATION METHODOLOGY

A. DATA

K-EmoPhone [7] contains 77 participants over about one week with 5,582 ESM responses; 3,771 of them have at least 50 % wrist-accelerometer coverage in the preceding 30 minutes and at least five complete minutes, and form our response set (`data/processed/kemophone_2hz_block.parquet`, from which the following are computed: 10 to 101 responses per user, median 49; period 1.4 to 6.5 days, median 6.3). Arousal and valence are reported on a scale centred at zero; we binarise at zero (above zero positive; 38.8 % positive for arousal, 58.0 % for valence) and test the user-median alternative in Section VI-A. The accelerometer is a wrist band at about 8 Hz, not the phone's own sensor; the data set does not release phone IMU data. All movement results in this paper are therefore results about wrist movement features that the phone computes identically, and the transfer to phone-worn movement is a limitation we return to in Section VII.

B. LOOP SIMULATOR

We replay each user in time order (`experiments/loop_sim/sim.py`; design and assumptions in `docs/LOOP_SIM_README.md`). For every response we compute the 30-minute summary at each of the 30 preceding minutes, exactly as the phone does; the summary at the response time equals the population model's input to the last digit (`experiments/loop_sim/build_minutes.py`, max |diff| = 0). The teacher labels each of these minutes with the response's own label flipped with probability $1-p$, present with probability d; bias b makes flips of negatives more likely than flips of positives while keeping the mean flip rate. The student is updated at every clock hour of the user's period, 100 steps per hour whether or not new minutes arrived, as on the device; over a six-day period this is about 14,000 steps on at most 2,000 highly correlated minutes whose labels copy about 49 responses. The prediction for a response is made before that response's teacher minutes are added, so no label information leaks forward. The first 60 % of each user's period is the learning phase; responses in the last 40 % are the evaluation set (minimum 10 per user; 65 users for arousal, 59 for valence). Conditions: (a) population model; (b) seed fine-tuned on previous ESM answers with the same learner; (b′) L2 logistic on previous ESM answers; (c) population model with per-user Platt scaling; (d) student on teacher labels; (d′) student with an L2 anchor towards the seed ($\lambda \in \{0.1, 0.3, 1\}$); (d″) intercept-only update with the weights frozen at the seed; (e) student with the guard, which starts disabled as on the device; (f) oracle teacher (p=1, d=1); and the untouched seed as a reference. Grid: $p \in \{0.6,0.7,0.8,0.9\}$, $d \in \{0.2,0.5\}$, $b \in \{0,0.2\}$, three seeds, 51 cells. The simulator cannot create affective states between responses; what it isolates is the effect of label noise, density and the on-device learning rule given the same underlying information.

C. STATISTICS

Within-user AUROC is averaged over users with at least 10 (simulator) or 20 (benchmark) evaluation responses and both classes. Confidence intervals are participant bootstraps (1,000 resamples of users); paired comparisons use the Wilcoxon signed-rank test with effect size $r = Z/\sqrt{n}$. We also report AUPRC, MCC at 0.5, Brier score and expected calibration error (`experiments/common/eval.py`). The guard analysis is a Bernoulli simulation with 10,000 runs per condition (`experiments/guard_sim/guard_sim.py`).

VI. RESULTS

A. POPULATION MODEL AND ABLATIONS (RQ1)

Table 2 compares the population model with baselines under leave-one-subject-out evaluation. Movement features beat the operating system's activity categories by 0.055 within-user AUROC for arousal (95 % CI 0.027 to 0.081, Wilcoxon $p = 4 \times 10^{-5}$) and by 0.056 for valence (0.024 to 0.094, $p = 0.002$), which confirms hypothesis H1a against that baseline. The comparison also sets a limit on what the features contribute: collapsing the wrist signal to three per-second categories (still, light, walking) and summarising those gives the same within-user AUROC as the 32 features (arousal 0.636 vs 0.623, difference +0.013 [−0.003, 0.033]). Any one of the three feature groups (magnitude, abruptness, stillness) alone matches the full set, and a 33-parameter L2 logistic model matches LightGBM (−0.005 [−0.020, 0.008]), whereas a 16-unit MLP is worse (−0.035 [−0.056, −0.016]). Time of day alone is at chance (0.526) and adds nothing to the movement model. The constant per-user mean label, which encodes only each user's level, reaches a pooled AUROC of 0.731 for arousal and 0.778 for valence, far above the movement model's 0.623 and 0.587, which illustrates why pooled AUROC is the wrong primary metric for personalisation. Binarising at the user median instead of zero changes within-user AUROC by less than 0.01.

TABLE 2. Within-user AUROC (mean over 72 / 71 users with ≥ 20 responses), participant-bootstrap CIs. Source: `results/benchmark/summary.csv`.

Model	Arousal within-user [95 % CI]	Δ vs movement	Valence within-user	Δ vs movement
Movement 32 features, LightGBM 50×3	0.623 [0.594, 0.650]	–	0.597 [0.562, 0.628]	–
OS activity categories	0.567 [0.541, 0.594]	−0.055 [−0.081, −0.027]	0.540 [0.505, 0.573]	−0.056 [−0.094, −0.024]
Wrist 3-class coarse	0.636 [0.609, 0.660]	+0.013 [−0.003, +0.033]	0.592 [0.557, 0.625]	−0.005 [−0.019, +0.010]
Hour of day	0.526 [0.496, 0.555]	−0.097 [−0.130, −0.064]	0.520 [0.489, 0.551]	−0.076 [−0.118, −0.036]
Movement, logistic (33 params)	0.617 [0.582, 0.648]	−0.005 [−0.020, +0.008]	0.586 [0.549, 0.620]	−0.010 [−0.027, +0.006]
Movement, MLP (16 hidden)	0.587 [0.562, 0.611]	−0.035 [−0.056, −0.016]	0.565 [0.531, 0.594]	−0.032 [−0.061, −0.004]

Window length (H1b, Figure 1): a 1-minute summary is worse than 30 minutes for arousal (−0.027 [−0.050, −0.005]) and 5 to 10-minute summaries are worse for valence (−0.060 [−0.092, −0.026] at 5 min), while 15, 30 and 60 minutes are indistinguishable. Thirty minutes is therefore a safe choice rather than an optimum, and the earlier claim that 60 minutes degrades accuracy is not supported by the confidence intervals.

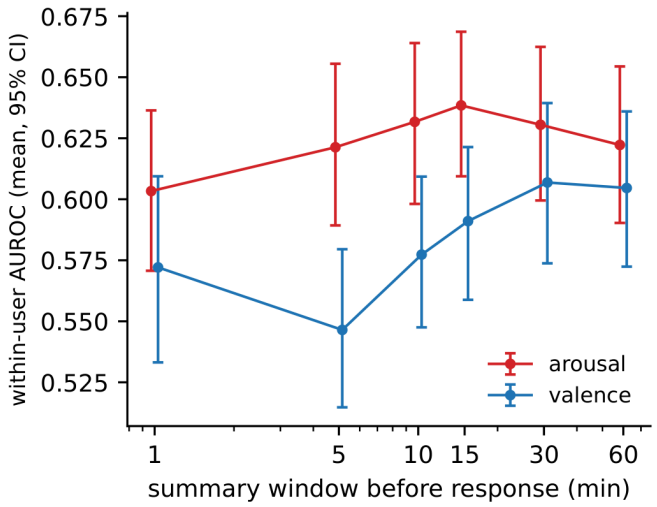


FIGURE 1. Summary window length and within-user AUROC of the population model (mean over users, participant-bootstrap 95 % CI; leave-one-subject-out on K-EmoPhone). One minute is worse than 30 minutes for arousal and 5 to 10 minutes are worse for valence; 15, 30 and 60 minutes are indistinguishable. Source: results/benchmark/summary.csv.

B. DENSE NOISY TEACHER VERSUS SPARSE ACCURATE LABELS (RQ2)

Figure 2 maps the pre-registered comparison: the difference in within-user AUROC between the self-trained student (dense, noisy) and the seed fine-tuned on previous ESM answers (sparse, accurate) over the (p, d) grid. The hypothesis H2, that the student wins above some teacher accuracy p^*, is not supported: the three-seed mean difference is at or below zero in all 32 cells (−0.001 to −0.023 for arousal, −0.009 to −0.064 for valence), no cell is significantly positive (upper confidence bounds reach +0.08 for arousal, so equivalence is not shown either), and the four valence cells at low accuracy and density ($p \leq 0.7$, $d = 0.2$) have seed-0 bootstrap intervals excluding zero on the negative side, none surviving a Holm correction over 32 tests. The stronger statement holds without qualification: in every arousal cell the student is below the population model with an upper confidence bound of −0.016 or less. The student's within-user AUROC is almost insensitive to p (arousal 0.51 to 0.54 from $p = 0.6$ to 0.9) and slightly better at the lower density for arousal (the reverse for valence), which already indicates that label noise is not the binding constraint.

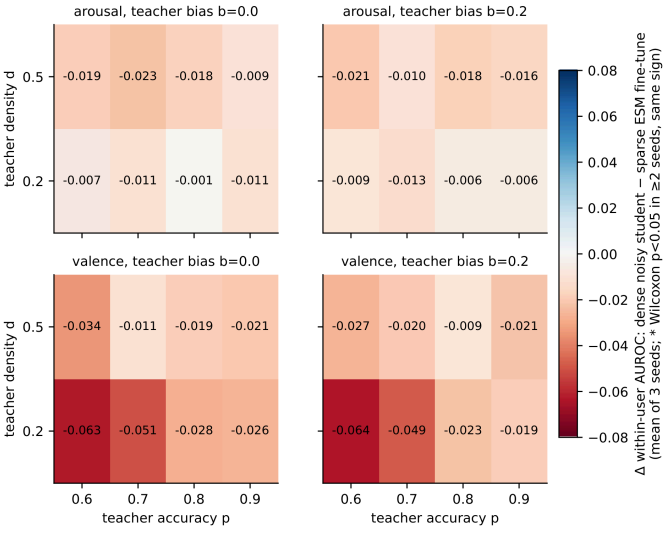


FIGURE 2. Pre-registered comparison (H2): difference in within-user AUROC between the self-trained student (dense, noisy teacher) and the seed fine-tuned on previous ESM answers (sparse, accurate) over teacher accuracy p and density d, for bias b = 0 and b = 0.2 (mean of three seeds). No cell is positive. Source: results/loop_sim/summary.csv.

The oracle condition, a noise-free label-copy teacher, makes the constraint explicit (Table 3 and Figure 3). With perfect labels on every minute, the student reaches 0.514 (arousal) and 0.567 (valence) within-user AUROC, below the population model (0.638, 0.627) and below its own untouched seed (0.626, 0.606). ESM fine-tuning with the same learner is at 0.536 and 0.579, the regularised ESM logistic at 0.582 and 0.578, and per-user Platt scaling at 0.537 and 0.527 (the slope is estimated negative for some users from ten labels). Every on-device update we tried lowers within-user discrimination, and the loss appears on the second day, when training starts, and does not recover within the week (Figure 4). Over the same conditions pooled AUROC rises from 0.617 to between 0.634 and 0.732 for arousal and from 0.586 to between 0.668 and 0.768 for valence: the student learns the user's level, which moves that user's probabilities up or down as a block and improves ranking across users without improving ranking within the user. Anchoring the weights to the seed with an L2 penalty halves the loss (−0.118 to −0.057 for arousal at $\lambda = 0.3$; all three λ give the same picture) but does not remove it. Freezing the weights at the seed and updating only the intercept, the most conservative learner we can build, still lowers within-user AUROC (0.603 for arousal, −0.034 [−0.056, −0.012]; 0.588 for valence) while raising pooled AUROC to 0.714 / 0.760: an hourly-varying level correction changes the order of a user's own responses across time, so even the level must be applied slowly or kept out of the ranking. The guard, which starts with the student disabled and enables it when the student leads by 0.05 on the recent probes, keeps the student active for 52 to 60 % of evaluation responses with 0.6 to 0.7 switches per user and also halves the loss (−0.059 for arousal, −0.041 [−0.091, +0.007] for valence).

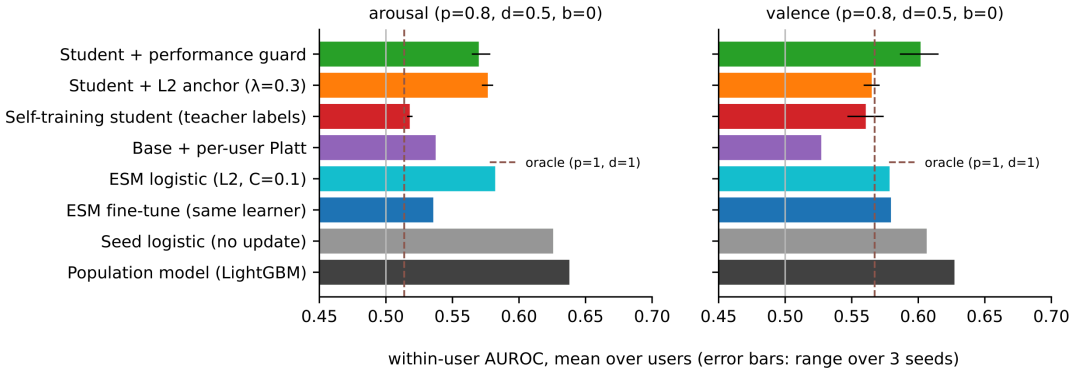


FIGURE 3. Within-user AUROC by condition at p = 0.8, d = 0.5, b = 0 (mean over users; error bars: range over three seeds; dashed line: oracle student with p = 1, d = 1). Every updated model is below the population model and the untouched seed. Source: results/loop_sim/summary.csv.

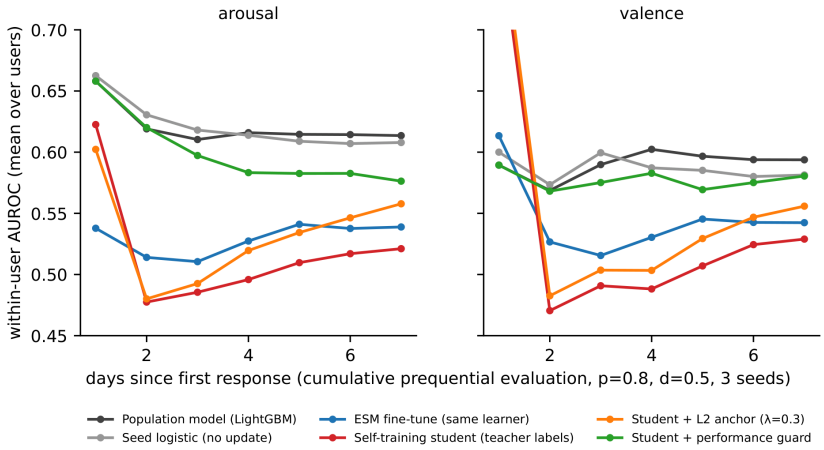


FIGURE 4. Cumulative prequential within-user AUROC by day since each user's first response (p = 0.8, d = 0.5, three seeds). The updated models fall on day 2, when training starts, and do not recover within the week. Source: results/loop_sim/preds_*.parquet.

TABLE 3. Loop simulator at $p = 0.8$, $d = 0.5$, $b = 0$ (within-user and pooled values are three-seed means; CIs from seed 0; 65 / 59 users). The population model's 0.638 differs from Table 2's 0.623 because it is scored on the last 40 % of each user's responses with the 10-response filter. Between 20 % and 37 % of users are above the population model under any updated condition (column `frac_users_gt_base`). Source: `results/loop_sim/summary.csv`.

Condition	Arousal within-user	Δ vs base [95 % CI]	Valence within-user	Δ vs base	Pooled AUROC (ar./val.)
Population model (LightGBM, fixed)	0.638	–	0.627	–	0.617 / 0.586
Seed logistic, no update	0.626	−0.012 [−0.033, +0.006]	0.606	−0.021 [−0.052, +0.008]	0.604 / 0.558
ESM fine-tune, same learner	0.536	−0.102 [−0.150, −0.051]	0.579	−0.048 [−0.109, +0.008]	0.634 / 0.672
ESM logistic, L2	0.582	−0.056 [−0.092, −0.015]	0.578	−0.049 [−0.110, +0.013]	0.650 / 0.668
Base + per-user Platt	0.537	−0.100 [−0.137, −0.063]	0.527	−0.100 [−0.159, −0.036]	0.732 / 0.757
Student, teacher p=0.8, d=0.5	0.518	−0.118 [−0.173, −0.061]	0.561	−0.080 [−0.138, −0.020]	0.647 / 0.703
Student + L2 anchor λ=0.3	0.577	−0.057 [−0.098, −0.014]	0.565	−0.068 [−0.115, −0.021]	0.719 / 0.765
Student + guard	0.570	−0.059 [−0.104, −0.016]	0.602	−0.041 [−0.091, +0.007]	0.662 / 0.715
Student, intercept-only update	0.603	−0.034 [−0.056, −0.012]	0.588	−0.039 [−0.068, −0.012]	0.714 / 0.760
Oracle student (p=1, d=1)	0.514	−0.124 [−0.177, −0.071]	0.567	−0.060 [−0.113, −0.008]	0.655 / 0.718

What the simulator can and cannot say. Because the teacher's labels are derived from the response labels, the dense signal carries no information that the sparse signal lacks; the simulator therefore tests the noise and density trade-off *at equal information*, and the learning rule's behaviour, not the possibility that expression captures affective states between responses. Within that scope the result is unambiguous: at K-EmoPhone's volume (about 49 responses and 650 teacher minutes per user) the hourly unregularised update over-fits the user's level, and a better teacher cannot fix that. Section VII draws the design consequence.

C. GUARD BEHAVIOUR (RQ3)

Table 4 and Figure 5 give the behaviour of four guards over 28 simulated days with three random probes per day, a base accuracy of 0.60 and a student whose accuracy differs by Δ (results at base accuracy 0.55 and 0.65 differ by at most 0.02). The student starts enabled, the worst case for a bad student; on the device and in the loop simulator it starts disabled. The two models' errors are drawn independently; correlated errors, which are likely in practice, would make every guard somewhat more powerful. The current fixed-threshold rule detects a student that is 0.10 worse within 14 days with

probability 0.76 and a student 0.20 worse with probability 0.94, with a median delay of 4.7 days, but it also reverts a student that is exactly as good as the base 48 % of the time within 14 days (68 % within 28 days) and switches about once per month at $\Delta = 0$. The Bayesian posterior and paired-CI guards cut the false-revert rate to 8 to 11 % at the price of power 0.25 to 0.30 and a median delay of 13 days; a sequential probability ratio test on discordant probes at $\alpha = \beta = 0.05$ almost never decides within the window. Forty-two probes cannot separate accuracies 0.10 apart reliably, so the choice is which error to accept. Because a false revert only returns the user to the population model, which Section VI-B shows is the better model at this data volume, the current rule's asymmetry is defensible for the revert direction; the re-enable rule (+0.05) is the one that triggers on noise, and a stricter or Bayesian re-enable criterion would be the first change we would make. The pre-registered prediction was half right: false reverts above 0.2 were predicted, but power below 0.5 was not.

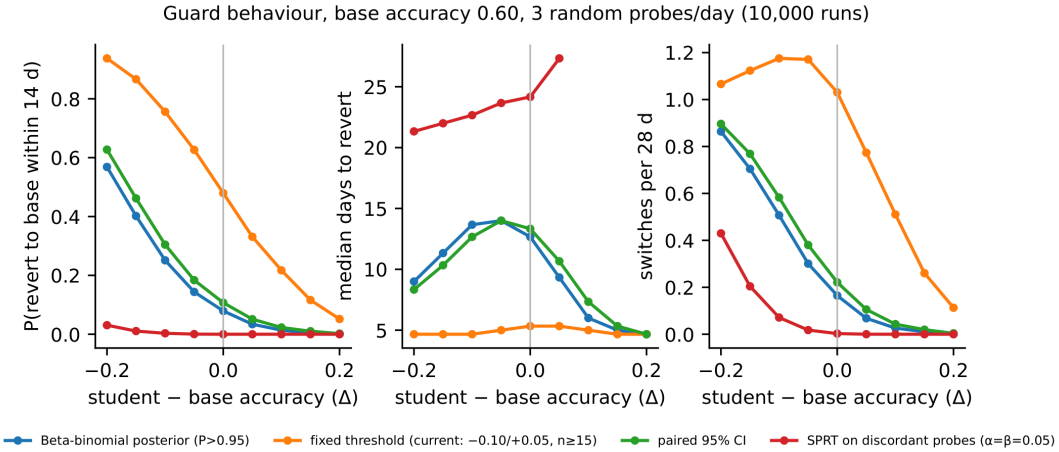


FIGURE 5. Guard behaviour against the student-minus-base accuracy difference Δ: probability of reverting to the population model within 14 days, median days to revert, and switches per 28 days (base accuracy 0.60, three random probes per day, 10,000 runs). Source: results/guard_sim/guard_sim.csv.

TABLE 4. Guard simulation, base accuracy 0.60, 10,000 runs. Source: `results/guard_sim/guard_sim.csv`.

Guard	Power, $\Delta = -0.10$, 14 d	Power, $\Delta = -0.20$	False revert, $\Delta = 0$, 14 d	28 d	Median delay	Switches / 28 d at $\Delta = 0$
Fixed threshold (current)	0.76	0.94	0.48	0.68	4.7 d	1.03
SPRT on discordant probes	0.00	0.03	0.00	0.00	22.7 d	0.00
Beta-binomial posterior ($P > 0.95$)	0.25	0.57	0.08	0.15	13.7 d	0.16
Paired 95 % CI	0.30	0.63	0.11	0.21	12.7 d	0.22

D. SELECTION BIAS OF MODEL-TRIGGERED PROMPTS (RQ4)

We reproduce the arm-A rule on the replayed data by applying the population model to the 30-minute summary at every minute before a response and flagging the response when the estimate moves at least 0.25 away from the estimate at the previous response. The rule fires for 8.5 % of arousal windows (322 of 3,771) and 2.4 % of valence windows. Flagged responses are more extreme (mean $|p - 0.5|$ 0.116 against 0.085, a factor 1.36), their pooled AUROC is within 0.002 of the random-time value, and only four users accumulate the ten flagged responses needed for a within-user estimate. Lowering the trigger to 0.15 flags a third of the windows and lowers pooled AUROC by 0.022; a quantile-based trigger at the 75th percentile lowers it by 0.030 with a within-user difference of −0.016 [−0.063, +0.029]. Reweighting the flagged responses by the inverse of each user's trigger rate (inverse-probability weighting) gives pooled AUROC of 0.631, 0.603 and 0.597 for the three triggers, so the weighting does not recover the random-time value either. A per-minute change rule, the naive reading of the specification, almost never fires with a 30-minute summary (median minute-to-minute change 0.036, maximum 0.26; `results/loop_sim/rq4_selection_consecutive.csv`, `docs/ASSUMPTIONS.md` A8). Model-triggered prompts therefore produce a shifted and small sample whose bias changes sign with the trigger definition; evaluating or guarding on them would be unreliable, which is the justification for the random arm (Figure 6).

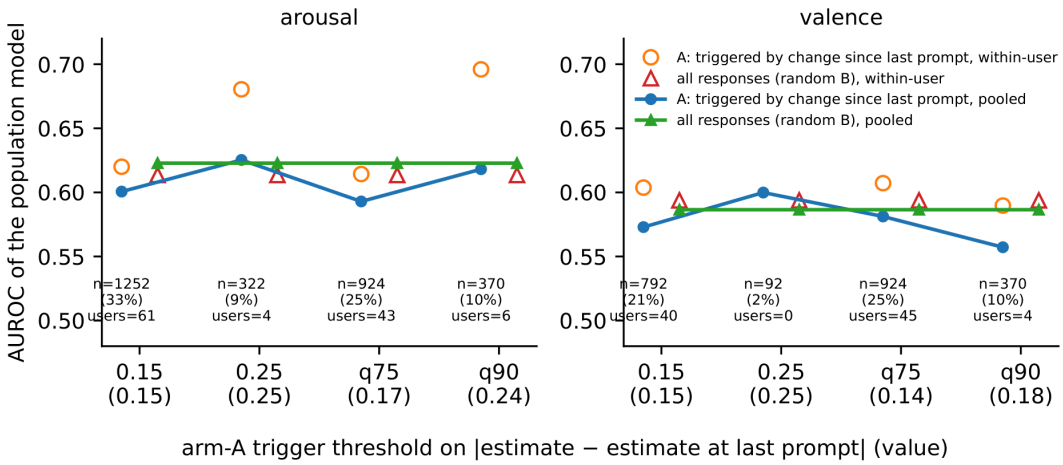


FIGURE 6. Selection bias of model-triggered prompts (arm A): pooled and within-user AUROC of the population model on responses flagged by the arm-A rule against all responses, for four trigger thresholds (value in parentheses); n, fraction flagged and number of users with at least ten flagged responses are printed. Source: results/loop_sim/rq4_selection.csv.

E. PRIVACY (RQ6)

Feature re-identification. With the 32 summary features of a single 30-minute window and a time-ordered split, a logistic attacker identifies the user among 77 with top-1 accuracy 0.070 (5.4 times chance) and top-5 0.193; majority voting over ten windows reaches 0.186. Non-linear attackers do not do better (kNN 0.027, LightGBM 0.052), so the identifying information is a small linear component of the feature levels. The absolute risk grows in small candidate sets: among 20 users top-1 is 0.167 and among 10 users 0.224 with top-5 0.719. Rounding features to one decimal reduces the 77-way rate to 0.052 (4.0 times chance) at no measurable accuracy cost (within-user AUROC −0.001 [−0.009, +0.006] for arousal, −0.010 [−0.023, +0.004] for valence), whereas rounding to integers reaches 0.037 (2.8 times chance) but collapses accuracy to 0.55 and 0.52 (`results/privacy/quant_cost.csv`).

Weight leakage. The 33 student weights are a far stronger identifier than the features. Registering each user's weights from two simulator seeds and querying with the third (the seeds differ only in the teacher's random flips), nearest-neighbour matching identifies the user with top-1 0.818 (63 times chance) and top-5 0.961; removing the shared seed component leaves the rate unchanged, and the anchored student is even more identifiable (0.987) because it is less perturbed by label noise. Gaussian noise with the same standard deviation as the weights is needed to halve the rate (0.416). A design that exports student weights, for aggregation or federated training, would therefore leak identity far more readily than the features do; our system keeps the weights on the device and this result is the reason.

F. SYSTEM COST (RQ7)

On the Galaxy Note9, three runs of the probe (64 samples, 100 steps) took 22.7 to 44.4 ms for training (0.23 to 0.44 ms per step), 2.3 to 34.8 ms to load the model, 0.6 ms for eight inferences, and 0.9 to 1.3 MB of additional memory; the probabilities agreed with the Python reference to a maximum difference of 6.0×10^{-8}. On the phone's own log of 330 minutes, computing one minute's 16 features and the 30-minute summary took a median 1.6 ms (95th percentile 2.6 ms). On a PC with a single-threaded LiteRT interpreter the hourly update scales as expected with parameters and batch: 33 parameters take 0.4 / 1.5 / 4.3 ms at batch 64 / 640 / 2,000, 330 parameters 1.2 / 10.1 / 46.5 ms, and 3,300 parameters 10.5 / 140 / 1,472 ms. The Note9 ran the same 33-parameter, 64-sample probe 60 to 120 times slower than the PC interpreter (22.7 to 44.4 ms against 0.4 ms), so a model one hundred times larger than ours at the full 2,000-sample buffer would take roughly 1.5 to 3 minutes per hour on the phone: feasible, but no longer negligible. On the device the hourly trainer has so far executed only on the probe batch: in the author's two-day log the teacher produced 31 face-detected minutes, below the 50-sample minimum, so every logged `train` event had zero samples, and the end-to-end loop on real teacher labels remains to be exercised in the field study. Battery cost of the camera teacher was not measured (no device during this run; procedure in `docs/DEVICE_BENCH.md`), and an earlier batched-versus-continuous sensing comparison found no measurable difference because the delivered sensor rate was controlled by other applications (`results/S12_NOTES.md`).

VII. DISCUSSION

When does on-device self-training help? Not at one week and 49 self-reports per user, with this learner, whatever the teacher's accuracy. The simulator shows three things in sequence. First, the information that an individual's labels add at this volume is their *level*, and the level is absorbed by any update rule, which is why pooled AUROC improves under every condition. Second, the hourly full-batch update without regularisation over-fits that level and destroys within-user ordering, which is why within-user AUROC falls even with an oracle teacher; the seed, left untouched, is as good as the population model. Third, no amount of label accuracy repairs this, because the teacher we can simulate does not add information between responses. The conditions under which a camera teacher could help are therefore (i) a teacher that carries information the self-reports do not, which only a field study with expression and probes can establish; (ii) a learner that cannot absorb the level without also learning the ordering, for instance an anchored update whose level component is applied as a slowly varying calibration rather than trusted as discrimination (Table 3 shows that even an hourly intercept-only update costs 0.03 within-user AUROC); and (iii) a data volume larger than one week, which Taylor et al. and Meegahapola et al. had and we did not. We regard the first as the open question and

the second as an engineering change that the simulator can now evaluate before any deployment.

Level versus change. The practical consequence for applications is that a personalised model should be read as two outputs: a level, which it learns quickly and which is useful for calibration and thresholds, and a within-person ordering, which it inherits from the population model and which the user's own labels do not improve for a long time. Reporting a single AUROC hides this; we recommend reporting both, as Fisher et al. argue on other grounds.

Guard. With three random probes a day, the guard cannot both detect a 0.10 loss within two weeks and avoid reverting a good student; the asymmetry of the current rule (fast revert, easy re-enable) is right for the first half and wrong for the second.

Teacher validity (RQ5). The author's own 2-day log contained 360 minutes, 31 with a detected face, and three manual probe answers, none within five minutes of a face-detected minute (`results/teacher/teacher_validity_log.txt`), so the expression-to-affect rule remains unvalidated. Public face-plus-self-report data sets that could validate it (K-EmoCon, Emognition, AMIGOS) require access requests and are listed with procedures in `docs/DATASETS.md`.

Threats to validity. Wrist rather than phone accelerometer; a single binarisation of a bipolar scale (robust to the user-median alternative); the simulator's copy-label teacher; one week of data; a 2018 device; Korean student participants in the public data set. The selection-bias result depends on the trigger definition and the guard result on independence between the two models' errors, which in practice are correlated and would make all guards somewhat more powerful.

VIII. ETHICS AND PRIVACY

The camera is used only while the screen is on and the phone is held, at 2 frames per second, and only 52 blendshape coefficients per frame are retained; no image leaves the camera pipeline. All raw and derived data stay on the device, the app has no network permission, and exports are manual copies by the researcher. Section VI-E quantifies the two residual risks: summary features identify a user at 5.4 times chance in a closed set of 77 (more in small groups), and student weights identify at 63 times chance, so weights are never exported. The field study package (protocol, consent form, data management plan, pre-registration) is in `docs/field_study/`; a power analysis from the simulator's effect sizes puts the sample at 18 participants (Wilcoxon, power 0.8) over four weeks to detect a within-user AUROC difference of 0.10 if the per-user AUROC variance shrinks with the response count as assumed (`docs/ASSUMPTIONS.md` A10), and at 50 without that adjustment; a difference of 0.05 would need 67 to 188 participants (`results/field/power.csv`). The device log also records the foreground application package and media titles, which the consent form discloses as an optional item. No human-subject data beyond the author's own log were collected in this work.

IX. CONCLUSION

We built the complete on-device loop for cross-modal self-training of a movement-based affect estimator and asked, with a replay simulator and pre-registered hypotheses, when it helps. At one week of data it does not: every on-device update, including one with perfect labels, trades within-user discrimination for level calibration, and the gains that appear in pooled accuracy are the level. This is a precondition, not a verdict: it tells a designer what a camera teacher must add (information between self-reports), how the learner must change (keep the level as calibration, protect the ordering), how many random probes a guard needs, and why the student's weights must stay on the phone. The next step is the field study specified in this repository, whose first job is to measure whether expression carries the information that the public data cannot show.

REFERENCES

1. TensorFlow Lite Team, "On-device training in TensorFlow Lite," TensorFlow Blog, 2021. [Online]. Available: https://blog.tensorflow.org/2021/11/on-device-training-in-tensorflow-lite.html
2. H. Cai, C. Gan, L. Zhu, and S. Han, "TinyTL: Reduce Memory, Not Parameters for Efficient On-Device Learning," in *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*, 2020. [Online]. Available: https://proceedings.neurips.cc/paper/2020/hash/81f7acabd411274fcf65ce2070ed568a-Abstract.html
3. Y. D. Kwon, J. Chauhan, H. Jia, S. I. Venieris, and C. Mascolo, "LifeLearner: Hardware-Aware Meta Continual Learning System for Embedded Computing Platforms," in *Proceedings of the 21st ACM Conference on Embedded Networked Sensor Systems (SenSys '23)*, 2023, pp. 138–151, doi: 10.1145/3625687.3625804.
4. H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282. [Online]. Available: https://proceedings.mlr.press/v54/mcmahan17a.html
5. A. Z. Tan, H. Yu, L. Cui, and Q. Yang, "Towards Personalized Federated Learning," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 12, pp. 9587–9603, 2023, doi: 10.1109/TNNLS.2022.3160699.
6. L. Meegahapola et al., "Generalization and Personalization of Mobile Sensing-Based Mood Inference Models: An Analysis of College Students in Eight Countries," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 4, pp. 176:1–176:32, 2022, doi: 10.1145/3569483.
7. S. Kang et al., "K-EmoPhone: A Mobile and Wearable Dataset with In-Situ Emotion, Stress, and Attention Labels," *Scientific Data*, vol. 10, Art. no. 351, 2023, doi: 10.1038/s41597-023-02248-2.
8. P. Zhang et al., "EmoPhone: A Multi-Wave Dataset for In-the-Wild Mobile and Wearable Affect Sensing," 2026, *arXiv:2609.16581*. [Online]. Available: https://arxiv.org/abs/2609.16581
9. R. Wampfler, S. Klingler, B. Solenthaler, V. R. Schinazi, M. Gross, and C. Holz, "Affective State Prediction from Smartphone Touch and Sensor Data in the Wild," in *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, 2022, pp. 1–14, doi: 10.1145/3491102.3501835.
10. S. Woll, J. Müther, D. Birkenmaier, G. Biri, U. W. Ebner-Priemer, and M. Giurgiu, "Predicting Momentary Mood in Daily Life from Accelerometer Data: Evaluating Single vs. Multiple Sensor Locations Using Machine Learning," *Sensors*, vol. 26, no. 12, Art. no. 3688, 2026, doi: 10.3390/s26123688.
11. S. Taylor, N. Jaques, E. Nosakhare, A. Sano, and R. Picard, "Personalized Multitask Learning for Predicting Tomorrow's Mood, Stress, and Health," *IEEE Transactions on Affective Computing*, vol. 11, no. 2, pp. 200–213, 2020, doi: 10.1109/TAFFC.2017.2784832.
12. A. J. Fisher, J. D. Medaglia, and B. F. Jeronimus, "Lack of group-to-individual generalizability is a threat to human subjects research," *Proceedings of the National Academy of Sciences*, vol. 115, no. 27, pp. E6106–E6115, 2018, doi: 10.1073/pnas.1711978115.
13. S. Albanie, A. Nagrani, A. Vedaldi, and A. Zisserman, "Emotion Recognition in Speech using Cross-Modal Transfer in the Wild," in *Proceedings of the 26th ACM International Conference on Multimedia (MM '18)*, 2018, pp. 292–301, doi: 10.1145/3240508.3240578.
14. J. Ni, R. Sarbajna, Y. Liu, A. H. H. Ngu, and Y. Yan, "Cross-Modal Knowledge Distillation for Vision-to-Sensor Action Recognition," in *ICASSP*

2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing, 2022, pp. 4448–4452, doi: 10.1109/ICASSP43922.2022.9746752.

15. C. I. Tang, I. Perez-Pozuelo, D. Spathis, S. Brage, N. Wareham, and C. Mascolo, "SelfHAR: Improving Human Activity Recognition through Self-training with Unlabeled Data," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 1, pp. 36:1–36:30, 2021, doi: 10.1145/3448112.
16. M. S. H. Onim, A. M. Kiselica, and H. Thapliyal, "Emotion Detection in Older Adults Using Physiological Signals from Wearable Sensors," 2025, *arXiv:2507.08167*. [Online]. Available: https://arxiv.org/abs/2507.08167
17. H. Yu and A. Sano, "Semi-Supervised Learning for Wearable-based Momentary Stress Detection in the Wild," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 7, no. 2, pp. 83:1–83:23, 2023, doi: 10.1145/3596246.
18. Q. Xie, M. T. Luong, E. Hovy, and Q. V. Le, "Self-Training with Noisy Student Improves ImageNet Classification," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 10684–10695, doi: 10.1109/CVPR42600.2020.01070.
19. A. Blum and T. Mitchell, "Combining Labeled and Unlabeled Data with Co-Training," in *Proceedings of the Eleventh Annual Conference on Computational Learning Theory (COLT '98)*, 1998, pp. 92–100, doi: 10.1145/279943.279962.
20. D. Angluin and P. Laird, "Learning from Noisy Examples," *Machine Learning*, vol. 2, no. 4, pp. 343–370, 1988, doi: 10.1007/BF00116829.
21. B. Frénay and M. Verleysen, "Classification in the Presence of Label Noise: A Survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 25, no. 5, pp. 845–869, 2014, doi: 10.1109/TNNLS.2013.2292894.
22. N. van Berkel, D. Ferreira, and V. Kostakos, "The Experience Sampling Method on Mobile Devices," *ACM Computing Surveys*, vol. 50, no. 6, pp. 93:1–93:40, 2017, doi: 10.1145/3123988.
23. G. Eisele et al., "The Effects of Sampling Frequency and Questionnaire Length on Perceived Burden, Compliance, and Careless Responding in Experience Sampling Data in a Student Population," *Assessment*, vol. 29, no. 2, pp. 136–151, 2022, doi: 10.1177/1073191120957102.
24. T. Reiter and R. Schoedel, "Never miss a beep: Using mobile sensing to investigate (non-)compliance in experience sampling studies," *Behavior Research Methods*, vol. 56, no. 4, pp. 4038–4060, 2024, doi: 10.3758/s13428-023-02252-9.
25. N. Lathia, K. K. Rachuri, C. Mascolo, and P. J. Rentfrow, "Contextual Dissonance: Design Bias in Sensor-Based Experience Sampling Methods," in *Proceedings of the 2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '13)*, 2013, pp. 183–192, doi: 10.1145/2493432.2493452.
26. D. G. Horvitz and D. J. Thompson, "A Generalization of Sampling Without Replacement from a Finite Universe," *Journal of the American Statistical Association*, vol. 47, no. 260, pp. 663–685, 1952, doi: 10.1080/01621459.1952.10483446.
27. P. Klasnja et al., "Microrandomized Trials: An Experimental Design for Developing Just-in-Time Adaptive Interventions," *Health Psychology*, vol. 34, no. Suppl, pp. 1220–1228, 2015, doi: 10.1037/hea0000305.
28. L. Na, C. Yang, C. C. Lo, F. Zhao, Y. Fukuoka, and A. Aswani, "Feasibility of Reidentifying Individuals in Large National Physical Activity Data Sets From Which Protected Health Information Has Been Removed With Use of Machine Learning," *JAMA Network Open*, vol. 1, no. 8, Art. no. e186040, 2018, doi: 10.1001/jamanetworkopen.2018.6040.
29. N. Neverova et al., "Learning Human Identity From Motion Patterns," *IEEE Access*, vol. 4, pp. 1810–1820, 2016, doi: 10.1109/ACCESS.2016.2557846.
30. J. L. Kröger, P. Raschke, and T. R. Bhuiyan, "Privacy Implications of Accelerometer Data: A Review of Possible Inferences," in *Proceedings of the 3rd International Conference on Cryptography, Security and Privacy (ICCSP 2019)*, 2019, pp. 81–87, doi: 10.1145/3309074.3309076.
31. M. Malekzadeh, R. G. Clegg, A. Cavallaro, and H. Haddadi, "Mobile Sensor Data Anonymization," in *Proceedings of the International Conference on Internet of Things Design and Implementation (IoTDI '19)*, 2019, pp. 49–58, doi: 10.1145/3302505.3310068.
32. R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership Inference Attacks Against Machine Learning Models," in *2017 IEEE Symposium on Security and Privacy (SP)*, 2017, pp. 3–18, doi: 10.1109/SP.2017.41.
33. L. Melis, C. Song, E. De Cristofaro, and V. Shmatikov, "Exploiting Unintended Feature Leakage in Collaborative Learning," in *2019 IEEE Symposium on Security and Privacy (SP)*, 2019, pp. 691–706, doi: 10.1109/SP.2019.00029.
34. M. Abadi et al., "Deep Learning with Differential Privacy," in *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS '16)*, 2016, pp. 308–318, doi: 10.1145/2976749.2978318.
35. M. Benouis, E. André, and Y. S. Can, "Balancing Between Privacy and Utility for Affect Recognition Using Multitask Learning in Differential Privacy-Added Federated Learning Settings: Quantitative Study," *JMIR Mental Health*, vol. 11, Art. no. e60003, 2024, doi: 10.2196/60003.

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